



TSA National Convention

Washington DC:
June 2026

“Phobia”
Video Game Design

PHOBIA

Team ID: 999PR

About:

What is Phobia?

What is *Phobia*? In simplistic terms, *Phobia* is a retro 2D platformer with modern graphics and storytelling; however, that's not the full picture. *Phobia* has classic platforming features (platforms, double jump, dash, etc.) to evoke a sense of familiarity, but also modern graphics to further Elizabeth's story. Additionally, we made each level one of Elizabeth's fears, such as Nyctophobia (fear of the dark), Claustrophobia (fear of tight spaces), and Anthropophobia (fear of people), which creates unique level design and furthers the story.

What's the purpose?

The purpose of *Phobia* is to blend classic 2D platforming and pixel art with modern graphics and storytelling. In this sense, it's a retro game with a modern twist. We wanted the players of *Phobia* to experience a meaningful story about the perspective and self-hate of our protagonist, Elizabeth. To further emphasize our story, we made all the music and sound effects ourselves to create the exact feeling and vibe. We wanted players to not only have a good gameplay experience, but also a good storytelling experience too.

What's the target audience?

The target audience of *Phobia* is the majority of people who understand how classic pixel art platformers work, but want to see a more unique game. We keep the familiarity by incorporating basic platforming that our audience understands, but we quickly stand out from other retro games with modern lighting, normal maps, 2D shadows, bloom, and other post-processing. Although our main focus was the graphics, we still wanted some familiarity with our audience, which is why we chose to render the game at a pixel-by-pixel scale.

Controls:

Phobia's controls are simply those of a basic 2D platformer and introduce the player to each of the controls to ease them into the gameplay. Each section provides more details on the specifics of the *Phobia's* platforming.

Basic Controls Overview:

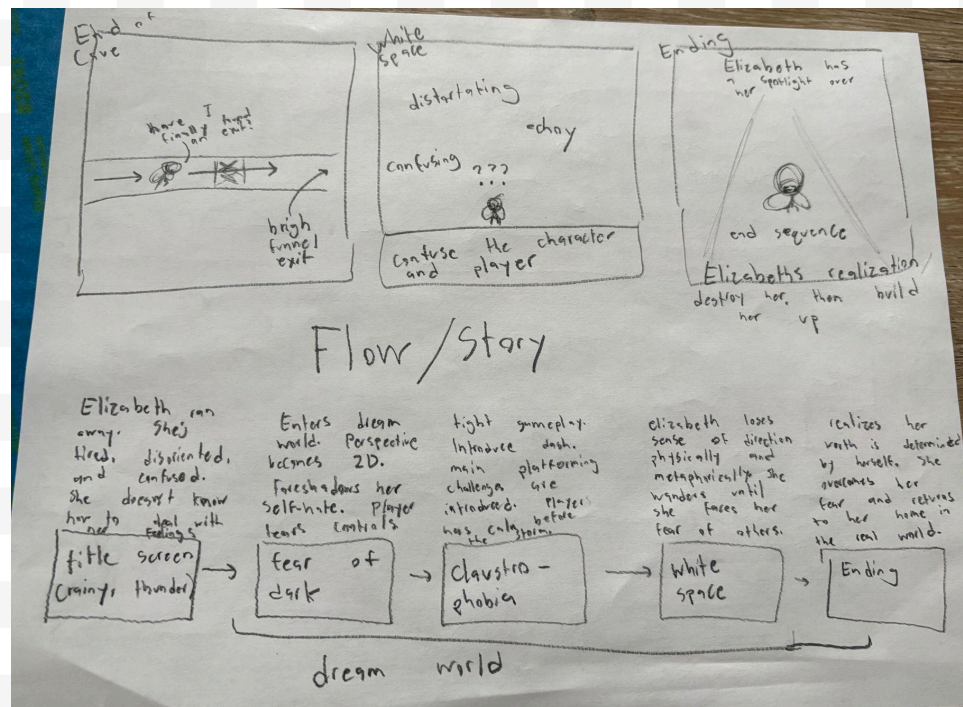
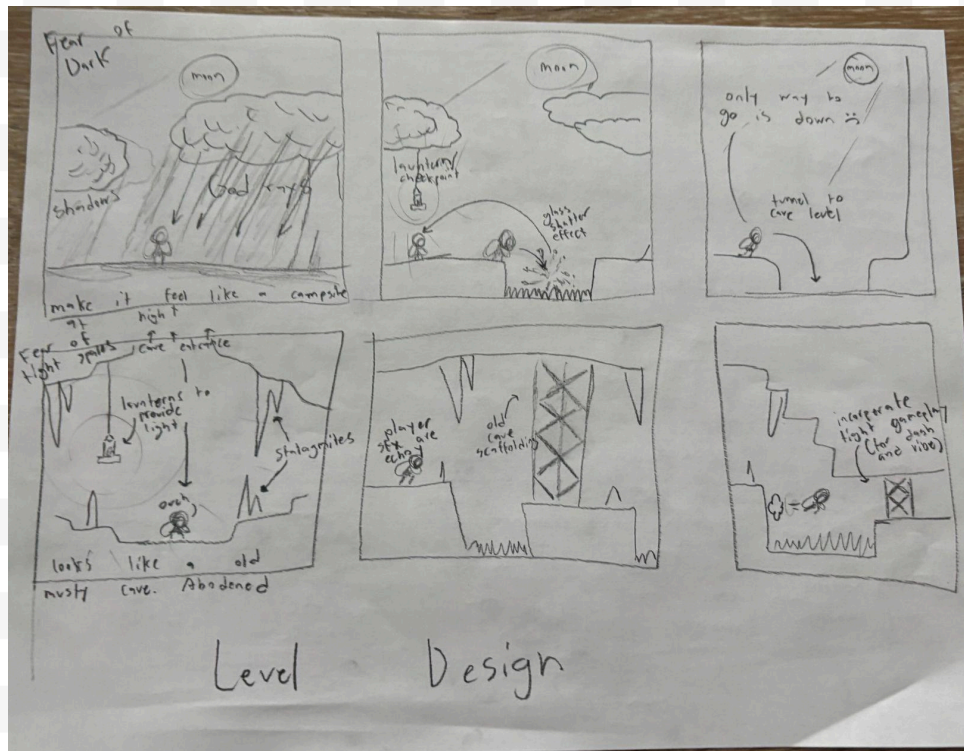
- **Up Arrow Key/W:** Jump *or* Double Jump (jump x2)
- **Left Arrow Key/A:** Move Left
- **Right Arrow Key/D:** Move Right
- **Space:** Start the game *or* Dash (In the Direction you're facing)

Up Arrow/W to jump is self-explanatory, although the game gives coyote time (or a 0.1-second delay) from when the player's hitbox leaves a platform to give a more responsive jump when leaving a ledge. Additionally, there is a 0.1-second jump buffer when you land, so you have 0.1 seconds before you land to initiate another jump immediately when you land for more precise platforming. Lastly, the height of a jump is dependent on how long you press the jump key. Jumping again in the air while Double Jump is unlocked will activate a second jump sequence while in the air.

The **left and right arrow (or A/D) keys are used to move left and right**. These can also be used to initiate coyote time (Ex, running off a ledge). Horizontal movement runs through a rigidbody object to calculate when the player collides frames before the actual contact. This approach allows for more accurate hitbox interaction, which is essential for our precision platformer.

Space is used to Dash and to start the game. On the title screen, space can be used to start the game after the 2-second-long intro sequence. To dash, simply hold Space in the air after unlocking the dash ability. While dashing, the gravity scale of the player is set to 0, and the player will move perfectly horizontally. On top of that, how long you hold Space determines how long the Dash will be. You can initiate a Double Jump while dashing. This will immediately cancel your Dash and begin the Double Jump Sequence.

Story board:



Student Copyright Checklist and Copyright letters:

STUDENT COPYRIGHT CHECKLIST (for students to complete and advisors to verify)

STUDENT: Answer question 1 below.

- 1) Does your solution to the competitive event integrate any type of music and/or sound? ☒ YES ☐ NO
If NO, go to question 2.
If YES, is the music and/or sound copyrighted? ☐ YES ☒ NO
If YES, move to question 1A. If NO, move to question 1B.
1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission and if permission is granted, include the permission in your documentation.
1B) Is the music/sound royalty free, or did you create the music/sound yourself? If YES, cite the royalty free music/sound OR your original music/sound properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, Robert Lee (chapter advisor), have checked my student's solution and confirm that any use of music/sound is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no music/sound included.

STUDENT: Answer question 2 below.

- 2) Does your solution to the competitive event integrate any graphics/videos? ☒ YES ☐ NO
If NO, go to question 3.
If YES, is(are) the graphics/videos copyrighted, registered and/or trademarked? ☐ YES ☒ NO
If YES, move to question 2A. If NO, move to question 2B.
2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for graphic/video used? If YES, move to question 3. If NO, ask for permission and if permission is granted, include the permission in your documentation.
2B) Is(are) the graphics/videos royalty free, or did you create your own graphic? If YES, cite the royalty free graphics/videos OR your own original graphics/videos properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of graphics/videos in his/her competitive event solution. Even if your student answers "NO" to question 2, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, Robert Lee (chapter advisor), have checked my student's solution and confirm that the use of graphics/videos with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no graphics/videos included.

STUDENT: Answer question 3 below.

- 3) Does your solution to the competitive event use another's thoughts or research? ☐ YES ☒ NO
If NO, this is the end of the checklist.
If YES, have you properly cited other's thoughts or research in your documentation? ☐ YES ☒ NO

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, Robert Lee (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of other's thoughts/research.

Student Name: TM, CB, DB

Chapter Advisor Signature: Robert W. Lee

Work Logs:

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	Task	Time involved	Team member responsible (student initials)	Comments
February 1-7th 1.	Player animations and tiles, squash and stretch and lighting implementaion	7 hours	TM	Added lots of animations and polish to character, also started the beginnings of the tile map and making the lighting look cool
February 7-14th 2.	Wind shader,and implementing sound effects	4 hours	TM	Added wind shader to the grass in the forest level, and also implemented sound effects and made the sound effects in ableton
February 14th-21 3.	Finish up lighting and visual polish, screen effects, lantern and parallax	7 hours	TM	Made all the parallax layers and made them accurate with the lgihting, also added the lanterns and their physics and got the forest level to be near complete visually
February 21-28 4.	Final visual polish, and optimization	5 hours	TM	Got the frame rate from 50-60 fps to around 110-120, and some things like god rays in the trees and also added correct lighting based on parallax layers, lots of math required here...
March1-6th 5.	Big final push and haul to get game done, cave tilesets and writing, make all the music	15 hours	TM	Made the music in ableton, and also made the parallax for the cave level, and the lighting for that, also worked on the final level story a little bit. Worked on Documentation portfolio and also made the video.
6.			TM	

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	Task	Time involved	Team member responsible (student initials)	Comments
February 10th, 2026 1.	Made concepts for the main character Elizabeth	2 hour	DB	Created concepts and finalized the character design for Elizabeth in the game
February 22nd, 2026 2.	Helped Start the UI Dialogue system	1 hour	DB	Helped Start the popup for UI Dialog for in game moments alongside Carter.
February 28th, 2026 3.	Added Tiles to the Forest level	3 hours	DB	Added the actual platforming near the end of the Forest level. Started with basic tiles to test out the platforming and added the grass and dirt tiles to replace them.
March 3rd, 2026 4.	Tile Map	2 hours	DB	Worked on the tile maps for the cave level
March 6th, 2026 5.	UI character icons	3 hours	DB	The previous UI character icons had very messy and sloppy hair flowing down or with no hair at all. I changed it to have good hair flow and style along with adding new emotions.
March 6th, 2026 6.	Disclosure Document	5 hours	DB	Added the purpose, description, controls, and the copyright document to the disclosure document

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	Task	Time involved	Team member responsible (student initials)	Comments
Jan 22-Feb 2 1.	Created basic gameplay	4 hours	CB	I created the basic player controller along with Coyote time, jump buffer, and dynamic jumping (higher gravity scale when falling)
Feb 3-Feb 11 2.	Added forest level (Fear of the dark) sprites	3 hours	CB	I created many of the basic sprites and normal maps, such as the tree sprite, and tile map.
Feb 12-Feb 23 3.	Lighting and Graphic changes	7 hours	CB	I added Fireflies, and lots lighting changes to make the game feel for developed. Most notably fully fixing the trees.
Feb 24-Feb 27 4.	Started cave level	5 hours	CB	Began cave level, created a specialized lantern to projects more into the cave, and fine tuned the parallax.
Feb 28-Mar 6 5.	Finished Cave level and game	15 hours	CB	Finished cave level, and Outro. Created a complete build of the game.
6.				

Allocated work log time:

TM: 38 hours
CB: 34 hours
DB: 16 hours